Sparse Coding-Inspired GAN for Hyperspectral Anomaly Detection in Weakly Supervised Learning

Yunsong Li, *Member, IEEE*, Tao Jiang, Weiying Xie, *Member, IEEE*, Jie Lei, *Member, IEEE*, and Qian Du, *Fellow, IEEE*

***Abstract*— Anomaly detection (AD) from hyperspectral images (HSIs) is of great importance in both space exploration and Earth observations. However, the challenges caused by insufficient datasets, no labels, and noise corruption substantially downgrade the accuracy of detection. To solve these problems, this article proposes a sparse coding (SC)-inspired generative adversarial network (GAN) for weakly supervised hyperspectral AD (HAD), named sparseHAD. It can learn a discriminative latent reconstruction with small errors for background pixels and large errors for anomalous ones. First, a background-category searching step is built to alleviate the difficulty of data annotation. Then, an SC-inspired regularized network is integrated into an end-to-end GAN to form a weakly supervised spectral mapping model consisting of two encoders, a decoder, and a discriminator. This model not only makes the network more robust and interpretable experimentally and theoretically but also develops a new SC-inspired path for HAD. Subsequently, the proposed sparseHAD detects anomalies in a latent space rather than the original space, which also contributes to its noise robustness. Quantitative assessments and experiments over real HSIs demonstrate the unique promise of the proposed sparseHAD. The code, data, and trained models are available at https://github.com/JiangThea/HAD.**

***Index Terms*— Anomaly detection (AD), generative adversarial network (GAN), hyperspectral images (HSIs), sparse coding (SC), weakly supervised learning (WSL).**

I. Introduction

ANOMALY detection (AD) tackles the problem of finding unknown space or material species without prior knowledge or samples, which has been studied in many data processing tasks [1]–[2]. Due to the rare and unbounded nature of anomalies, such a problem stimulates the use of representation theory. For instance, AD based on sparse coding (SC) [4], [5] has successfully implanted its potential in AD [6], [7], including the interpretation of brain neuron activity, the prevention of overfitting, and enhancement of network robustness [8].

However, aforementioned AD in traditional color images has its inherent limitations. Detectors tend to confuse in some challenging scenarios, e.g., noise interference, background clutter, object camouflage, and similar appearance of different materials [9]. In these cases, however, the intrinsic information of an anomaly and its surrounding background is distinguishable, and the most useful type of images capable of capturing the intrinsic information is the hyperspectral image (HSI) [10].

HSI usually appears in three dimensions, one dimension of which provides continuous spectral reflectance vectors containing essential attributes of materials, and the other two dimensions show spatial information. Thus, HSI is regarded as an advanced technique for Earth observation and has been widely concerned in environmental monitoring, resource exploration, fine agriculture, and management [11]. To fully investigate these applications, various data analysis methods have been derived, including land cover classification [12], target detection [13], and AD [14]. Among them, hyperspectral AD (HAD) aims to locate anomalous pixels that differ spectrally from background categories.

Although HSIs are capable of performing material recognition, due to the inherent factors of spectral imaging and rarity of anomalies, how to use this information effectively for HAD still faces many challenges. In particular, the challenges caused by **insufficient datasets**, **no labels**, and **noise corruption** substantially downgrade the accuracy of detection [15]. Therefore, in this article, we develop a weakly supervised perspective in the absence of prior knowledge and propose a novel AD framework termed **Sparse** Coding-inspired generative adversarial network (GAN) for **H**yperspectral **A**nomaly **D**etection **(SparseHAD)**. In this work, the problem of **insufficient datasets** in deep learning-based AD is the first concern. In HSI processing, affected by equipment and time cost, typically, only one HSI is involved in each image processing process. Direct input of a 3-D HSI into a deep learning network cannot make the network learn HSI features. Due to HAD aiming to detect pixels with dramatically spectral differences from their surroundings, it is reasonable to take spectral vectors as input. At this point, a $100 \times 100 \times 200$ HSI would have 100×100 samples with 200 dimensions.

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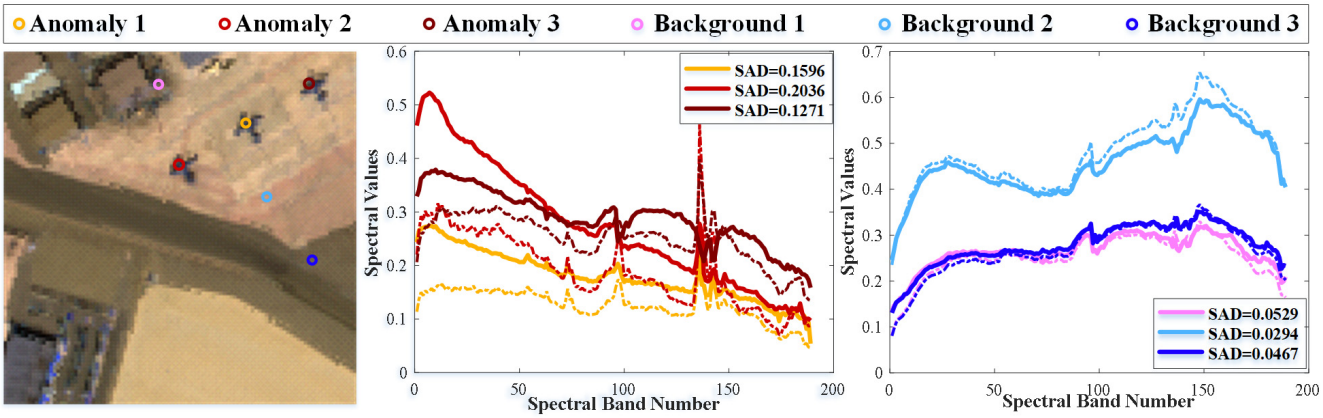


Fig. 1. Spectral reconstruction results and their spectral angle distance (SAD) values corresponding to different pixels obtained by a GAN model. (Left): Locations of anomaly pixels and background pixels in the color composites. (Middle) and (Right): Thick solid line and thin dotted line correspond to the spectral vector curves at the same location in the original HSI and the reconstructed HSI, respectively. The smaller the corresponding SAD values indicate that the spectral vectors are more similar, which also means that the reconstruction errors are smaller.

100×100 training samples would ensure the network learns rich spectral feature information.

Then the second problem is **no labeling**. Relying on the notion that the initially given labels of weakly supervised learning (WSL) are not always true [16], the WSL is introduced. A background-category searching step is built to eliminate potential anomaly samples and form a coarse background sample set. The embedding of this step not only breaks the limitation of the sample label of unsupervised learning but also matches the efficient detection performance of supervised learning [17]. Subsequently, the coarse background samples can be regarded as normal samples and then fed into the proposed weakly supervised spectral mapping model based on an end-to-end GAN. This model is expected to reconstruct the background spectral vectors as far as possible in the case that the anomalous spectral vectors cannot be well performed, as illustrated in Fig. 1.

Noise corruption is the driving force for the proposal of the SC-inspired regularized network (sparseNet), but the role of SC in the proposed model is far more than making the network more robust to noise. A deeper investigation shows that SC can assist the end-to-end GAN [with autoencoder (AE)] to avoid learning an extreme approximation of the identity function [18]. Furthermore, SC has strong interpretability, making our network robust both in theory and practice [19]. To make the optimization of attached sparseNet tractable, sparseNet can be recast into the above GAN, which is a data-driven SC-inspired network without the need for precomputed sparse codes. Accordingly, the proposed method can model a weakly supervised spectral mapping function and finally arrive at sparseHAD by exploring different reconstruction error calculation strategies: original space reconstruction (*Strategy 1*) and latent space reconstruction (*Strategy 2*). The latent reconstruction is preferred since it can eliminate the influence of image noise caused by degradation mechanisms, as shown in Fig. 2. Note that the strip noise can still be seen in *Middle*, while no trace of the strip noise can be seen in *Right*. This illustrates that latent reconstruction makes our model highly robust to noise because the latent space makes the network pay more attention to the difference from the intrinsic spectral attribute [2].

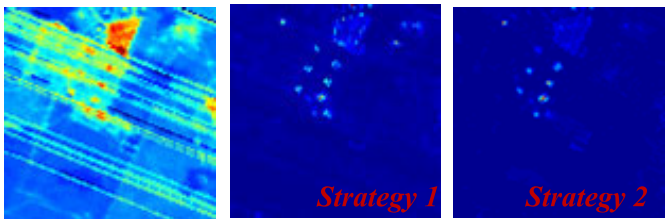


Fig. 2. Taking strip noise in HSIs as an example. (Left): Spatial map of the 190th band of the Texas-1 dataset. (Middle) and (Right): Detection maps on Texas-1.

Comparing with other existing AD algorithms in HSI, the main contributions of our weakly AD method lie in the followings: A novel interpretable and more robust sparesNet is developed, which can be efficiently implemented and seamlessly integrated into an end-to-end GAN. To the best of our knowledge, this is the first time that SC is introduced into HAD. Two new strategies are proposed to obtain the differential image, which are based on the original space and the latent space, respectively. The proposed framework predicts anomalies in a latent space rather than the original space. This makes the network more robust to noise, and avoids dimension disaster.

The remainder is organized as follows. Related work is presented in Section II. Section III briefly reviews the basic theories related to SC. The proposed sparseHAD is detailed in Section IV. Section V shows the assessments and generalization experiments. Conclusions are drawn in Section VI.

II. Related Work

HAD has attracted considerable interest from the research community over the decades because of its penchant for unknown targets. Based on the observations that anomalies in HSIs occur with a low probability, statistical methods are often applied. Reed Xiaoli (RX) [20] detector, as an important milestone in HAD, was introduced to distinguish anomalies from the background using their statistic differences. The algorithm assumes that the background of hyperspectral data obeys multivariate Gaussian distribution. Based on this statistical assumption, various improvements to the RX detector, such as local RX detector (LRX) [21], and

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kernel RX (KRX) [22], have been proposed to improve performance. Besides statistical models, representation-based methods including sparse representation (SRD) [23], collaborative representation-based detector (CRD) [24] and low-rank and sparse matrix decomposition-based anomaly detector (LSMAD) [25] are popular. CRD assumes that if a pixel belongs to the background, it can be represented by a linear approximation of its neighborhood pixels. SRD is a sparse version of CRD. LSMAD assumes that anomalies are sparsely distributed and that the background after anomaly removal has a low-rank attribute. By combining spectral unmixing, Qu *et al.* [26], [27] proposed a novel threefold detection algorithm based on dictionary-based low-rank decomposition. By considering the noise characteristics, Li *et al.* [14] recently designed a low-rank sparse matrix decomposition detector (LSDM_MOG) combining with a mixed noise model. The graph and total variation regularized low-rank representation-based detector (GTVLRR) [28] method first analyzes the local geometrical structure and spatial relationships, then introduces the regularization and total variation of graphs into the low-rank representation. Chang *et al.* [29] improved the traditional low-rank sparse decomposition method, GoDec, and proposed an orthogonal subspace-projection (OSP) version of GoDec. Zhang *et al.* [30] replaced the conventional low-rank regularization with spectral-difference low-rank regularization. The priority-based tensor approximation (PTA) method [31] introduces tensors into low-rank and sparse priors. Different from the previous work based on spectral difference, the detectors mainly based on spatial difference are also applied, such as attribute and edge-preserving filters-based detector (AED) [32] and structure tensor and guided filter-based detector (STGD) [33]. Both methods assume that the occurrence probability of an anomaly is much less than that of the background. Recently, other HAD methods have emerged. Fractional Fourier entropy-based detector (FrFE) [34] applied the fractional Fourier transform to represent signals in an intermediate-domain. Subpixel-pixel-superpixel guided fusion-based detector (SPSGF) [35] fully investigated the influence of the fusion of features at different pixel levels in HAD. However, with complex scenes and applications, the background of HSIs becomes complex, which limits the detection performance of traditional methods.

Deep learning-based methods can capture nonlinear and deep features, and realize complex function approximation, which has been widely concerned in different applications [36]. Considering the difficulty of obtaining labels for HAD data, most existing work on deep learning-based HAD can be roughly categorized twofold: semisupervised and unsupervised. Some semisupervised learning work, e.g., [37], [38], expose a common weakness that they are unable to acquire the expected background labels. As for unsupervised approaches, stacked AE [39] and adversarial AE are used to extract in-depth features in the latent layer [40], [41]. Hosseiny and Shah-Hosseini [42] realizes nonlinear feature extraction by the convolution stacked AE network. MC-AEN [43] combines global and local reconstruction errors of AE to achieve detection. Discriminative reconstruction constrained generative adversarial network for hyperspectral anomaly detection (HADGAN) [3] first employs GAN [44] to model background in HAD. Li *et al.* [45] uses a transferred convolutional neural network (CNN) to capture the difference between pixel pairs, while neural network and dictionary-based low-rank detector (NNDLR) [46] employs a CNN to extract the abundance maps. DeCNN-AD [47] integrates the prior denoising model based on CNN into the traditional low-rank decomposition. Jiang *et al.* [48] presented an integrated unsupervised low-rank embedded network (LREN) in which a Gaussian mixture model is embedded in the deep latent layer. Although these methods can produce good detections, the absence of labels limits their detection performance.

III. Preliminaries

SC can effectively discover semantic information from noisy and high-dimensional data [49]. Such advantages make SC favored by applications, such as image super-resolution [50] and video AD [51]. In particular, the SC technique for AD is to model normal samples so that normal samples can be reconstructed with small errors, while abnormal samples have large reconstruction errors [51].

Given a set of features corresponding to normal samples as $\boldsymbol{Y} = \{\boldsymbol{y}_i \in \mathbb{R}^p\}_{i=1}^m$, the objective of SC is to linearly reconstruct $\boldsymbol{Y}$ by an unknown coding dictionary matrix $\boldsymbol{A} \in \mathbb{R}^{p\times q}$ with small error $\boldsymbol{\varepsilon}$, i.e., $\boldsymbol{Y} = \boldsymbol{A}\boldsymbol{\alpha} + \boldsymbol{\varepsilon}$, where $\boldsymbol{\alpha} \in \mathbb{R}^{q\times m}$ is the unknown sparse coefficients. The nonconvex objective function is formulated as

$$\sum_{i=1}^{i=m} \|\boldsymbol{y}_i - \boldsymbol{A}\cdot\boldsymbol{\alpha}_i\|_2^2 + \sum_{i=1}^{i=m} \mathbf{S}(\boldsymbol{\alpha}_\mathbf{i}) \tag{1}$$

where the first term calculates the reconstruction error, evaluating how well the input sample is reconstructed by the dictionary, and the second term corresponds to a sparse term with $\mathbf{S}(\cdot)$ being a nonlinear sparsity penalty function. In the past practice, a common method to solve such problem is to alternatingly optimize $\boldsymbol{A}$ and $\boldsymbol{\alpha}$, which correspond to two optimization procedures: dictionary learning and sparse approximation. Then, a dictionary can be learned that encodes all normal samples. In the test stage, when a sample enters, its sparsity coefficient is first calculated according to dictionary $\boldsymbol{A}$, and then its type can be discriminated.

IV. Proposed Method

The proposed weakly supervised framework learns a discriminative latent reconstruction with small errors for background samples and large errors for anomaly samples, which means the latent differential image (i.e., latent reconstruction errors) can highlight the anomaly instances and restrain the background instances. Such a framework can be constructed by recasting SC as a novel regularized AE unit to GAN, termed sparseHAD. As shown in Fig. 3, it consists of three modules: **data preparation**, **adversarial learning**, and **inference**.

A 3-D cube HSI $\mathbf{H} = \{\boldsymbol{h}_i \in \mathbb{R}^L | i = (i_1, i_2) = N\cdot(i_1-1)+i_2\}_{i_1=1,i_2=1}^{i_1=M,i_2=N}$ with L dimensions can be reshaped into a 2-D matrix $\boldsymbol{H} = \{\boldsymbol{h}_i \in \mathbb{R}^L\}_{i=1}^{M\times N}$, where (i_1, i_2) represents the position coordinates of the ith spectral vector in the spectral domain, M and N are the height and width in the spatial dimension of HSI, respectively. Besides, a coarse

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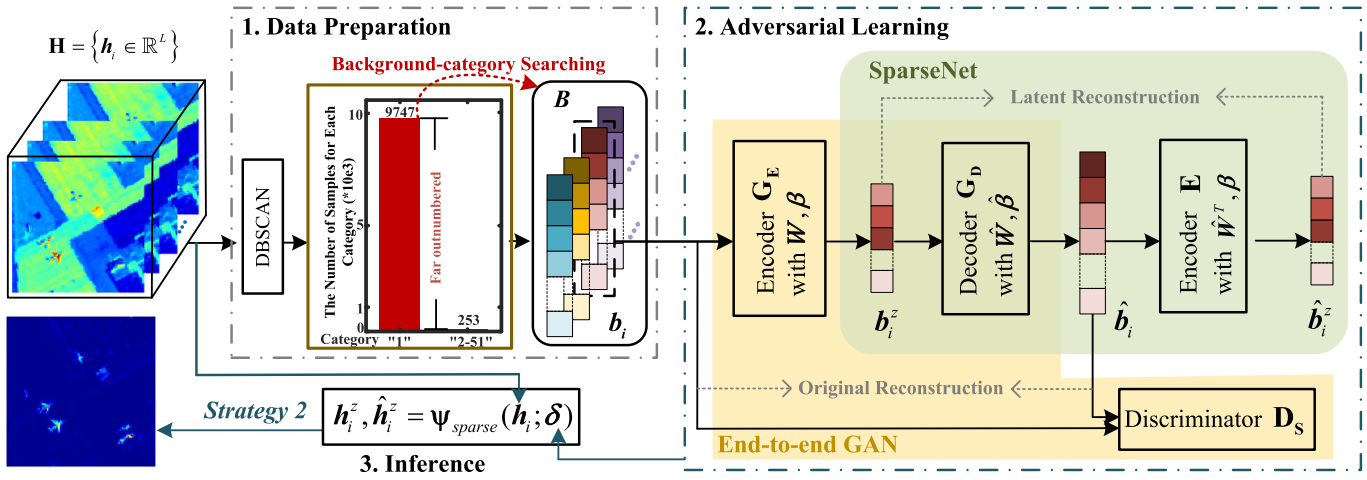


Fig. 3. High-level overview of the proposed sparseHAD method in HSI.

background samples set $\boldsymbol{B} = \{\boldsymbol{b_i} \in \mathbb{R}^L\}_{i=1}^{N_B}$ can be obtained by discarding the potential anomaly samples in set $\boldsymbol{H}$, where N_B is the number of coarse background samples. Consider the introduction of SC, the sparseHAD includes four variables, i.e., $\boldsymbol{H}$, $\boldsymbol{H}^Z$, $\hat{\boldsymbol{H}}$, and $\hat{\boldsymbol{H}}^Z$, where $\boldsymbol{H}$ and $\hat{\boldsymbol{H}}$ belong to the original HSI space, and $\boldsymbol{H}^Z$ and $\hat{\boldsymbol{H}}^Z$ belong to the latent HSI space. Meanwhile, $\hat{\boldsymbol{H}}$ is the reconstruction of $\boldsymbol{H}$, and $\boldsymbol{H}^Z$ and $\hat{\boldsymbol{H}}^Z$ are latent mappings of $\boldsymbol{H}$ and $\hat{\boldsymbol{H}}$, respectively. Based on these notation, a nonlinear spectral mapping model $\Psi_{\text{sparse}}(\cdot)$ from the original HSI space to the latent space is established as follows:

$$\boldsymbol{h}_i^z, \hat{\boldsymbol{h}}_i, \hat{\boldsymbol{h}}_i^z = \Psi_{\text{sparse}}(\boldsymbol{h}_i; \boldsymbol{B} | \boldsymbol{B} \subset \boldsymbol{H}) \tag{2}$$

where $\boldsymbol{h}_i^z \subset \boldsymbol{H}^Z \in \mathbb{R}^{(M\times N)\times L_Z}$, $\hat{\boldsymbol{h}}_i \subset \hat{\boldsymbol{H}} \in \mathbb{R}^{(M\times N)\times L}$, $\hat{\boldsymbol{h}}_i^z \subset \hat{\boldsymbol{H}}^Z \in \mathbb{R}^{(M\times N)\times L_Z}$, and $L_Z (\ll L)$ refers to the dimension of the latent space. The learning objective of the model $\Psi_{\text{sparse}}(\cdot)$ is to capture the discriminative feature between anomaly and background, resulting in the latent differential image $\mathcal{A}^{\mathcal{Z}}$ between $\boldsymbol{H}^Z$ and $\hat{\boldsymbol{H}}^Z$ more discriminative than the original differential image $\mathcal{A}$ between $\boldsymbol{H}$ and $\hat{\boldsymbol{H}}$.

A. Data Preparation

This step, also termed background-category searching, is to prepare for WSL and intends to predict coarse labels of the background samples in the given input. According to the fact that there is no prior knowledge in HAD, an unsupervised clustering method with two parameters (ϵ: the minimum distance between adjacent points, MinPts: the minimum number of points), named density-based spatial clustering of applications with noise (DBSCAN [52]), can be exploited.

DBSCAN is a density-based clustering algorithm, which generally assumes that the category can be determined by the density of the distribution of samples (i.e., sample similarity for HSI). For samples of the same category, the distance between them is relatively close (the sample similarity is high for HSI). In a HAD image, spectral vectors of anomalous samples are dramatically different from that of background samples. Thus, anomalous samples have low similarity with the background samples of different categories, which means that the interclass difference between the anomalous samples and the background samples is much greater than the intraclass difference between the background samples. Such data property enables the DBSCAN algorithm to classify the background category and anomalous category based on sample similarity. Due to atmospheric conditions and equipment errors, some noise categories may also be included. Considering the theoretical basis and statistical fact, a probability-based background searching step is adopted. It starts by calculating the sample number of each category in $\boldsymbol{C}$. Given an input HSI $\boldsymbol{H}$, the DBSCAN scans every pixel's neighbors within ϵ distance. Once the number of neighbors of the core pixel exceeds MinPts, a cluster is formed. Subsequently, the neighbors within ϵ distance of the core pixel are assembled iteratively to obtain the category probability map

$$\begin{aligned} \boldsymbol{C} &= \text{DBSCAN}(\boldsymbol{H}) \\ \boldsymbol{C} &= \{c_i | i = (i_1, i_2)\}_{i_1=1, i_2=1}^{i_1=M, i_2=N}. \end{aligned} \tag{3}$$

Then, calculating the sample number of each category in $\boldsymbol{C}$. For different HSIs, the ranges of values of c_i (i.e., the number of categories) for DBSCAN are different and can be as large as fifty. Experimentally, the number of samples in one category "1" (usually classified as category "1") in $\boldsymbol{C}$ is much higher than that of other categories, such as 9747 to 253. According to the statistical fact that the probability of background occurrence in an HSI is far higher than the anomaly, the number of background samples is larger than that of anomalous samples. Thus, after searching for samples of category "1," these samples can be classified into the background samples set $\boldsymbol{B}$ with the coarse label "1," while the remaining categories are discarded as potential anomaly samples as

$$\boldsymbol{B} = \{\boldsymbol{h}_i | i \leftarrow c_{(i_1,i_2)} = 1\}. \tag{4}$$

Note that the starting points for the clustering in DBSCAN are randomly selected. Because of the high probability of background samples, the starting point is usually the background sample. This makes category "1" usually the

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background category. After many tests, the background category in the data used is all category "1."

For coarse background samples, they are allowed to have some category errors, i.e., the predicted coarse labels of $\boldsymbol{B}$ are not always true. Consequently, the model $\Psi_{\text{sparse}}(\cdot)$ is learned on $\boldsymbol{B}$ in the form of WSL.

B. SC-Inspired Adversarial Learning

A novel SC-inspired GAN network, consisting of two encoders, a decoder, and a discriminator, is constructed to form three subnetworks, as shown in Fig. 3.

To meet the requirements of reconstruction and overcome the imbalance between GAN's generator and discriminator, the **first subnetwork**, namely **an end-to-end GAN with AE**, as [53] is constructed. The AE, consisting of an encoder ($\mathbf{G_E}$) and a decoder ($\mathbf{G_D}$), behaves as the generator ($\mathbf{G}$) in GAN, and plays a role of reconstructing. Model $\Psi_{\text{sparse}}(\cdot)$ on $\boldsymbol{B}$ is learned by the mapping $\boldsymbol{B} \to \boldsymbol{B}^{\boldsymbol{Z}} = \mathbf{G_E}(\boldsymbol{W}\boldsymbol{B} + \boldsymbol{\beta})$, and then by the mapping $\boldsymbol{B}^{\boldsymbol{Z}} \to \hat{\boldsymbol{B}} = \mathbf{G_D}(\hat{\boldsymbol{W}}\boldsymbol{B}^{\boldsymbol{Z}} + \hat{\boldsymbol{\beta}})$. Especially, $\boldsymbol{B}^{\boldsymbol{Z}} = \{\boldsymbol{b}_i^z \in \mathbb{R}^{L_Z}\}_{i=1}^{N_B}$ and $\hat{\boldsymbol{B}} = \{\hat{\boldsymbol{b}}_i \in \mathbb{R}^{L}\}_{i=1}^{N_B}$ correspond to the latent space and the original space, respectively. The spectral discriminator $\mathbf{D_S}$ can not only see through the $\mathbf{G}$'s tricks but also avoid blurring of the reconstructed image $\hat{\boldsymbol{B}}$ caused by AE [54].

The reconstructed encoder network E is the **second subnetwork** and attempts to compress $\hat{\boldsymbol{B}}$ into $\hat{\boldsymbol{B}}^{\boldsymbol{Z}}$ by $\hat{\boldsymbol{B}}^{\boldsymbol{Z}} = \mathbf{E}(\hat{\boldsymbol{B}})$, where $\hat{\boldsymbol{B}}^{\boldsymbol{Z}} = \{\hat{\boldsymbol{b}}_i^z \in \mathbb{R}^{L_Z}\}_{i=1}^{N_B}$. Compared to the previous methods (in which latent vectors are obtained by minimizing the distance between encoder and decoder), this subnetwork acquires latent vectors via minimizing latent distance. The subnet works this way is to mitigate the effects of noise (e.g., strip noise in HSIs) [2].

The third subnetwork is **a SC-inspired regularized network (SparseNet)** whose objective is to build networks from a neuroscience perspective to make them more robust [8] and conducive to AD [19]. Meanwhile, the end-to-end generator $\mathbf{G}$ may learn an approximation to the identity function without additional network constraints, which would make it impossible for the generator to identify any anomaly from the background [18]. Thus, a novel SparseNet is suggested to be attached.

Mathematically, SC is modeled as a generative model as neuroscientists [5] do. The most critical problem is how to apply SC to the GAN model. Based on the understanding of SC theory and the deconstruction of (1), the following principle is employed to build a sparse regularized network.

1) Eliminating the impact of precomputing. Most SC methods need to precompute $\boldsymbol{\alpha}$, which largely affects the performance of those methods [55]. However, GAN-based sparseNet directly recasts SC to GAN, avoiding the calculation of $\boldsymbol{A}$ and $\boldsymbol{\alpha}$.
2) Learning overcomplete representations. The advantages of overcomplete representation include stronger robustness in the case of unavoidable noise, more sparsity, greater flexibility in data matching structure, and compliance with some response characteristics of primary visual cortex neurons [56].

An overcomplete representation can be constructed when the number of basis vectors on an overcomplete basis is greater than the dimension of the input, and the input representation is not unique [56], that is, dimension q must be greater than dimension p in $\boldsymbol{A}$ for (1). Considering these two points and the GAN constructed above, the most appropriate means of applying SC to our model is taking the latent layer output (i.e., $\boldsymbol{B}^{\boldsymbol{Z}}$) of the end-to-end GAN model as the input of the sparseNet. Particularly, $\hat{\boldsymbol{W}}$ can be regarded as the dictionary $\boldsymbol{A}$ in the sparseNet. Meanwhile, the L1 normalization is chosen as the sparse penalty term. $\boldsymbol{s}$ is specified as $\hat{\boldsymbol{b}}_i$ for SC. Therefore, (1) can be reformulated as

$$\sum_{i=1}^{i=N_B} \left\|\boldsymbol{b}_i^z - \hat{\boldsymbol{W}} \cdot \hat{\boldsymbol{b}}_i\right\|_2^2 + \sum_{i=1}^{i=N_B} \mathbf{S}\left(\hat{\boldsymbol{b}}_i\right). \tag{5}$$

Then combined with encoder network $\mathbf{E}$, the forward feedback of sparse network can be expressed as

$$\begin{aligned} \hat{\boldsymbol{b}}_i &= \mathbf{G_D}\left(\boldsymbol{b}_i^z\right) \\ \hat{\boldsymbol{b}}_i^z &= \mathbf{E}\left(\hat{\boldsymbol{b}}_i\right). \end{aligned} \tag{6}$$

The learning objective is to fully mine the representative background information from the coarse background samples set $\boldsymbol{B}$ during the learning process. Therefore, when the anomaly samples pass through the well-learned network, there are large errors between the reconstructed anomaly samples and the input anomaly samples because the network parameters are not suitable for the anomaly samples. Moreover, if the above original reconstructed output does not retain anomalous information well, then the latent reconstructed output obtained from the original reconstruction encoded by the subnetwork $\mathbf{E}$ will correspondingly fail to retain the valid information of anomalies [2]. Such a situation would result in a large error between the latent layer and the latent reconstructed layer. To verify the efficacy of our idea, two losses (adversarial and sparsity) are incorporated into the learning objective, each of which optimizes the corresponding subnetworks and accounts for its contribution.

The adversarial loss of the end-to-end GAN is to reconstruct the background samples set $\boldsymbol{B} = \{\boldsymbol{b}_i\}_{i=1}^{N_B}$. Meanwhile, to capture the distribution of background samples more effectively, L1 normalization is applied to calculate the reconstruction error [2]. The adversarial loss with the reconstruction loss is reformulated as

$$\begin{aligned} L_A &= \min_{\mathbf{G}}\left(\alpha_d \max_{\mathbf{D_S}}(L_d) + \alpha_r L_r\right) \\ L_d &= \mathbb{E}_{\boldsymbol{b}_i \sim p(\boldsymbol{b}_i)}\left\|\mathbf{D_S}(\boldsymbol{b}_i) - \mathbb{E}_{\boldsymbol{b}_i \sim p(\boldsymbol{b}_i)}\mathbf{D_S}\left(\hat{\boldsymbol{b}}_i\right)\right\|_2 \\ L_r &= \mathbb{E}_{\boldsymbol{b}_i \sim p(\boldsymbol{b}_i)}\|\boldsymbol{b}_i - \hat{\boldsymbol{b}}_i\|_1, \boldsymbol{b}_i = \mathbf{G}(\boldsymbol{b}_i). \end{aligned} \tag{7}$$

The sparsity loss is obtained by integrating subnetworks SparseNet and $\mathbf{E}$. To avoid learning the parameters of another new encoder and to reduce the computational overhead, the decoder $\mathbf{G_D}$ parameters $\hat{\boldsymbol{W}}$ are reused in $\mathbf{E}$ as $\hat{\boldsymbol{W}}^T$, just like tied weighted AE. The sparse loss formula can be recast as follows:

$$\begin{aligned} L_S &= \mathbb{E}_{\boldsymbol{b}_i \sim p(\boldsymbol{b}_i)}\left(\|\boldsymbol{b}_i^z - \hat{\boldsymbol{W}} \cdot \hat{\boldsymbol{b}}_i\|_F^2 + \|\hat{\boldsymbol{b}}_i\|_1\right) \\ \boldsymbol{b}_i^z &= \mathbf{G_E}(\boldsymbol{b}_i). \end{aligned} \tag{8}$$

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TABLE I
DETAILS OF THE DATASETS USED IN THE EXPERIMENT

Datasets	Scene	Resolution	Sensor	Spatial size	Bands	Anomaly material	Anomaly proportion
G1	Airport	3.4m	AVIRIS	100×100	191	Three planes	0.6%
G2	Urban	3.5m	AVIRIS	100×100	191	unknown	0.52%
S1	Urban	3.5m	AVIRIS	100×100	189	Three planes	0.57%
S2	Airport	unknown	AVIRIS	100×100	189	Three planes	1.34%

*Note that AVIRIS means the Airborne Visible Infrared Imaging Spectrometer.

The final objective of the model $\Psi_{\text{sparse}}(\cdot)$ gives

$$L_F = L_A + L_S. \tag{9}$$

Model $\Psi_{\text{sparse}}(\cdot)$ is optimized by minimizing the loss function. When the multinetworks are jointly learned to a certain epoch, the parameters $\boldsymbol{\delta} = (\boldsymbol{W}, \boldsymbol{\beta}, \hat{\boldsymbol{W}}, \hat{\boldsymbol{\beta}})$ of model $\Psi_{\text{sparse}}(\cdot)$ are fixed and used to inference for AD.

C. Inference for AD

During the **inference process**, the given $\boldsymbol{H} = \{\boldsymbol{h}_i\}_{i=1}^{M\times N}$ is fed into the well-learned model as

$$\boldsymbol{h}_i^z, \hat{\boldsymbol{h}}_i, \hat{\boldsymbol{h}}_i^z = \Psi_{\text{sparse}}(\boldsymbol{h}_i; \boldsymbol{\delta}). \tag{10}$$

By the above inference, the model $\Psi_{\text{sparse}}(\cdot)$ learns three outputs. Due to only focusing on the learning backgrounds characteristics, this model can reconstruct the background samples well but cannot reconstruct the anomalies well. This results in small errors between the reconstructed background samples and the input background samples, and large errors between the reconstructed anomaly samples and the input anomaly samples. Aiming to finding the anomalies in the HSI, two differential images from (10) are acquired. They are the original differential image $\mathcal{A}$ and the latent differential image $\mathcal{A}^{\mathcal{Z}}$, respectively, expressed as

$$\begin{aligned} \text{Strategy1}: \ \mathcal{A} &= \|\hat{\boldsymbol{h}}_i - \boldsymbol{h}_i\|_1 \\ \text{Strategy2}: \ \mathcal{A}^{\mathcal{Z}} &= \|\hat{\boldsymbol{h}}_i^z - \boldsymbol{h}_i^z\|_1 \end{aligned} \tag{11}$$

where $\mathcal{A} = \{\boldsymbol{a}_i \in \mathbb{R}^L\}_{i=1}^{M\times N}$ and $\mathcal{A}^{\mathcal{Z}} = \{\hat{\boldsymbol{a}}_i \in \mathbb{R}^{L_Z}\}_{i=1}^{M\times N}$. Equation (11) describes two strategies for calculating the reconstruction error corresponding to Section I.

During the **detection process**, the Mahalanobis distance is adopted on $\mathcal{A}^{\mathcal{Z}}/\mathcal{A}$ to detect anomalies. Although both $\mathcal{A}$ and $\mathcal{A}^{\mathcal{Z}}$ can highlight the anomaly instances and restrain the background instances, $\mathcal{A}^{\mathcal{Z}}$ has better robustness than $\mathcal{A}$. Consistent with this, the Mahalanobis distances in $\mathcal{A}^{\mathcal{Z}}$ will get clearer anomalies and less background interference. The preferred final detection results are as follows:

$$\boldsymbol{D}_{\boldsymbol{a}} = (\hat{\boldsymbol{a}}_i - \boldsymbol{\mu})^T \boldsymbol{\Gamma}^{-1} (\hat{\boldsymbol{a}}_i - \boldsymbol{\mu}) \tag{12}$$

where $\boldsymbol{\mu} \in \mathbb{R}^{L\times 1}$ and $\boldsymbol{\Gamma} \in \mathbb{R}^{L\times L}$ represent the mean vector and covariance matrix of the matrix $\mathcal{A}^{\mathcal{Z}}$, respectively. In subsequent experiments, the effectiveness of the above statement will be assessed.

V. EXPERIMENTS

A. Experimental Setup

The datasets that are employed in evaluating the sparse-HAD method include Gulfport [32], Gainesville [32], San Diego-1 [25], and San Diego-2 [57], the details of which are listed in Table I. It is worth noting that the datasets contain both point and structural anomalies, and are in the form of different scales.

The evaluation metrics used to report experimental results are the receiver operating characteristic (ROC) curve [59] and the area under the ROC curve (AUC), where ROC can be plotted by the true-positive rate (P_d) and the false-positive rate (P_f) at various thresholds (τ). Furthermore, the closer the AUC of (P_d, P_f) value is to 1, the better the detection performance. Conversely, the closer the AUC of (P_f, τ) value is to 0, the lower the probability of false detection. In AD, the main goal is to find and locate anomalies. Therefore, the AUC of (P_d, P_f) value is taken as our primary consideration standard, followed by the AUC of (P_f, τ) value.

The baseline methods are six frequently cited as well as state-of-the-art HAD methods, including two deep learning-based methods spectral adversarial feature learning for anomaly detection (SAFL) [58], HADGAN [3], four traditional-based methods CRD [24], LSMAD [25], FrFE [34], and LSDM_MoG [14]. In particular, HADGAN is a GAN-based method. To verify the superiority of the network, only the RX detector is used after the HADGAN network, termed HADGAN_RX. For fine comparison, the SAFL network was reproduced as recommended in the article [58] and replaced the detector with the RX detector, termed SAFL_RX.

1) Implement Details: The proposed model is implemented in TensorFlow (v1.10.0, Python 3.6.0, and CUDA 10.0) by optimizing the networks using Adam [60] with an initial learning rate 0.0001, the number of hidden nodes 20. The batch size is the same as the number of input spatial pixels, and the epoch is 3000. The weighting parameters in (7) is chosen as $\boldsymbol{\alpha_d} = 0.5$ and $\boldsymbol{\alpha_r} = 50$ empirically. Both $\mathbf{G_E}$ and $\mathbf{G_D}$ adopt two fully connected network layers, which are then activated by leaky rectified linear unit (ReLU) function. Empirically, the number of latent layer units is set to 20. The $\mathbf{D_S}$ has the same structure as the $\mathbf{G_D}$. $\mathbf{E}$ and $\mathbf{G_E}$ share the same structural details but different parameters.

2) Parameter Analysis: In the data preparation stage, there are two parameters: ϵ and MinPts. Considering the pixel-level processing characteristics of the proposed method and its insensitivity to parameter ϵ, MinPts was fixed as 1. Therefore,

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TABLE II
PARAMETERS OF ALL METHODS OVER THE FOUR DATASETS

Datasets	SparseHAD	HADGAN_RX	SAFL_RX	FrFE	LSDM_MoG	CRD	LSMAD
	Fixed parameters						
All	MinPts=1	As [3]	As [58]	–	k=4	λ=10^{-6}	r=3,k=0.003
–	Settable parameters						
G1	ϵ=0.13	–	–	–	r=6	w_{out}=17,w_{in}=13	–
G2	ϵ=0.08	–	–	–	r=2	w_{out}=9,w_{in}=7	–
S1	ϵ=0.19	–	–	–	r=4	w_{out}=13,w_{in}=7	–
S2	ϵ=0.22	–	–	–	r=4	w_{out}=17,w_{in}=13	–

TABLE III
EVALUATION AUC SCORES OF THE MODEL'S EFFECTIVENESS ON DIFFERENT DATASETS

Datasets	The AUC scores of (P_d, P_f)				The AUC scores of (P_f, τ)			
	sparseHAD_L	_O	_EL	_EO	sparseHAD_L	_O	_EL	_EO
G1	**0.9886**	0.9690	0.9842	0.9700	0.0185	0.0203	**0.0088**	0.0204
G2	**0.9792**	0.9475	0.9678	0.9474	**0.0052**	0.0322	0.0074	0.0323
S1	**0.9846**	0.7598	0.9797	0.7598	**0.0101**	0.0273	0.0103	0.0273
S2	**0.9914**	0.9467	0.9895	0.9468	0.0140	0.0322	**0.0117**	0.0322
Average	**0.9860**	0.9060	0.9803	0.9060	0.0120	0.0280	**0.0096**	0.0280

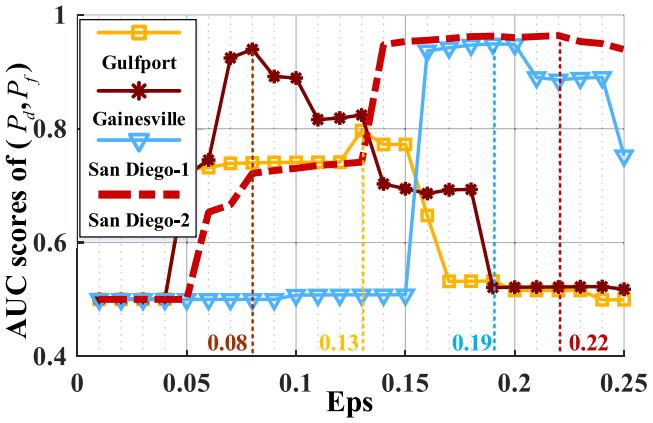


Fig. 4. Effects of the parameter Eps (ϵ) over the AUC scores of (P_d, P_f) on each dataset.

here focuses on the parameter ϵ, which mainly controls the outputs of the category probability map. With Eps varying from 0.01 to 0.25, the optimal ϵ values for different datasets are shown in Fig. 4. The AUC scores achieve the bests when ϵ is 0.13 for Gulfport, 0.08 for Gainesville, 0.19 for San Diego-1, and 0.22 for San Diego-2, respectively. For the baseline methods, It can be found the optimal parameters for them to achieve comparable performance. Their exact configurations are listed in Table II.

B. Experimental Analysis

1) Effectiveness Assessment: According to Section IV-C, there are two strategies to calculate the reconstruction error. As shown in Fig. 2, Strategy 2 is more robust to noise than Strategy 1. Furthermore, Strategy 2 is more resource-efficient, because $\mathcal{A}^{\mathcal{Z}}$'s dimension is much smaller than $\mathcal{A}$. In this section, the superiority of Strategy 2 will be proved quantitatively. First, sparseHAD_O is taken to denote the detection method corresponding to Strategy 1, and use sparseHAD_L to represent the detection method on Strategy 2. Then, to strengthen the argument, the new set of comparison experiments is added, in which **E** with the updatable parameters is used instead of **E** with the tied weighted $\mathbf{G_D}$'s parameters, which form two new comparison methods, i.e., sparseHAD_EO and sparseHAD_EL. The performance comparisons are reported in Table III. One can find that sparseHAD_L/sparseHAD_EL outperforms sparseHAD_O/sparseHAD_EO consistently on all evaluation metrics, which provides additional evidence for the superiority of Strategy 2. It can also be found that the AUC scores of (P_d, P_f) for sparseHAD_L are always better than sparseHAD_EL, which explains the advantages and rationality of using weighted AE in our network. Considering the above analysis, sparseHAD_L is adopted as sparseHAD.

2) Ablation Study: To justify the benefits of the proposed sparseHAD method, the following ablation studies are conducted: #1 denotes the proposed method, #2 means the end-to-end GAN without SC is adopted as the core network, #3 represents that only AE is used to obtain reconstruction error, and #4 means that HSIs are directly fed into the RX detector to calculate the Mahalanobis distances without prior access to any network. The AUC scores are tabulated in Table IV. The average AUC scores of (P_d, P_f) obtained for #4 is already high at 0.9400. Therefore, from this point, even the slightest improvement is important. When the deep neural network is introduced, the performance of the system improved by an average of 2.69%. That is to say, using the deep neural network is beneficial. By replacing AE with an end-to-end GAN, the performance was further improved by an average of 0.725%, which indicates that adversarial learning has a positive influence on HAD. Finally, by adding a

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TABLE IV
EVALUATION AUC SCORES OF THE ABLATION STUDY ON DIFFERENT DATASETS

Datasets	The AUC scores of (P_d, P_f)				The AUC scores of (P_f, τ)			
	#1	#2	#3	#4	#1	#2	#3	#4
G1	**0.9886**	0.9811	0.9715	0.9520	**0.0185**	0.0387	0.0243	0.0257
G2	**0.9792**	0.9658	0.9583	0.9511	**0.0052**	0.0305	0.0298	0.0360
S1	**0.9846**	0.9632	0.9578	0.9054	**0.0101**	0.0265	0.0301	0.0390
S2	**0.9914**	0.9865	0.9800	0.9515	**0.0140**	0.0323	0.0464	0.0713
Average	**0.9860**	0.9742	0.9669	0.9400	**0.0120**	0.0320	0.0327	0.0430

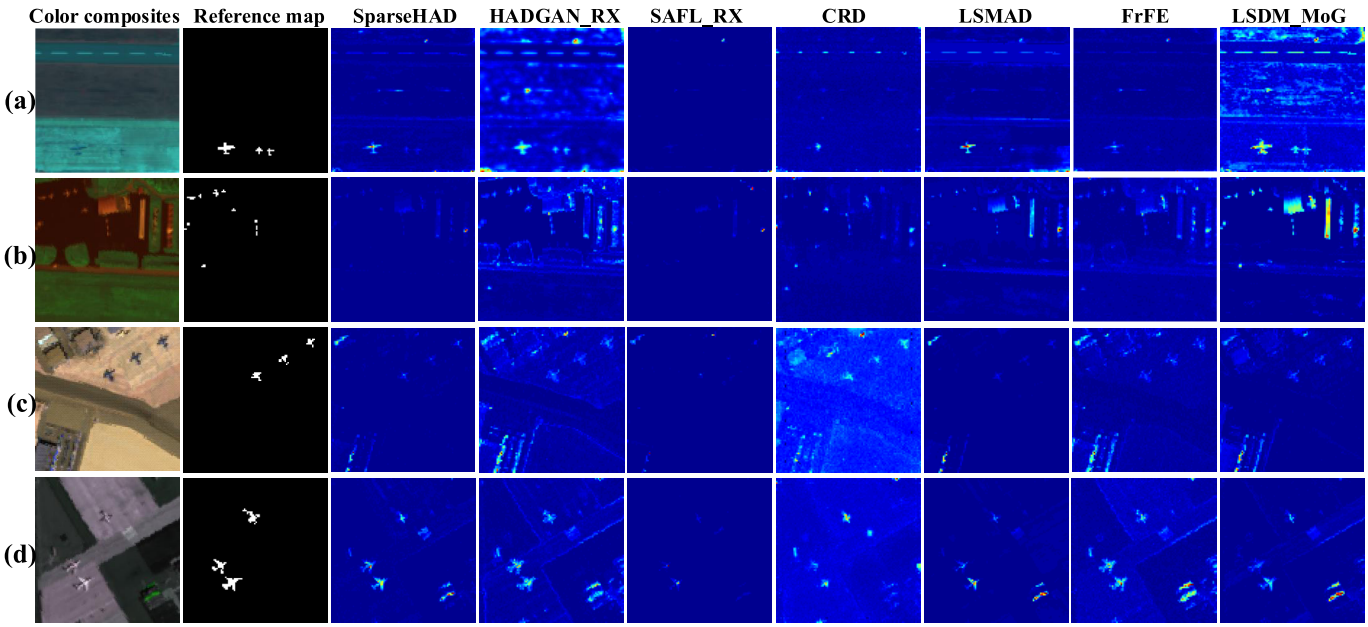


Fig. 5. Color composites, binary reference maps and detection maps of the compared methods for (a) Gulfport, (b) Gainesville, (c) San Diego-1, and (d) San Diego-2.

sparse regularized network, the performance improves further by 1.18%, indicating that the sparse regularized network is effective. By observing the AUC scores of (P_f, τ), we can also draw a conclusion consistent with the AUC scores of (P_d, P_f), which once again shows the effectiveness and superiority of the network designed.

3) Performance Comparison: Based on the parameter settings in Table II, all methods are guaranteed to achieve optimal performance. The visual detection maps in Fig. 5 reveal that the detection maps obtained by the proposed sparseHAD are closest to the ground truth. The sparseHAD method can ensure that anomalies with different sizes and shapes are highlighted while suppressing the background of rich categories. Other algorithms are more likely to ignore anomalies, get distracted by the background, or lose the shape of the anomaly. In particular, in Fig. 5(a), sparseHAD can capture the shapes of aircraft of different sizes as well as their positions. However, SAFL_RX, CRD, and FrFE do not. Although HADGAN_RX and LSDM_MoG can also detect anomalies clearly, they are greatly influenced by the background. The LSMAD almost lost two smaller aircraft targets. Taking Fig. 5(b) and (c), sparseHAD ensures that small targets are detected while background interference is minimized. In Fig. 5(d), the scenario of the anomalies with more structural information, our method still does a good job of detecting anomalies, even their edges. FrFE can obtain anomalies similar to those found in sparse HAD, but it is more susceptible to background interference. As discussed above, the HADGAN_RX method is always trapped in background interference. Although SAFL_RX is barely affected by its background in every dataset, its anomalous targets are rarely highlighted and lose most of their shape information. CRD is susceptible to window size and unable to obtain complete anomalous targets on all datasets. The FrFE and LSDM_MoG methods have poor robustness to overcome noise and background. Anomalies detected by the LSMAD method are always less complete than those detected by sparseHAD. The ROC curve of (P_d, P_f) is shown in Fig. 6. This visualization reveals that the proposed sparseHAD can produce a higher P_d as P_f varying from 10^{-4} to 1. The corresponding AUC results of the competing methods and the sparseHAD method are shown in Table V, where sparseHAD by itself delivers the best performance in terms of AUC scores of (P_d, P_f). Specifically, with regard to the AUC score of (P_d, P_f), it exceeds the second-best FrFE by an average of 0.98%. Although its AUC scores of (P_f, τ) are higher than LSDM_MoG, it still shows a great competitive

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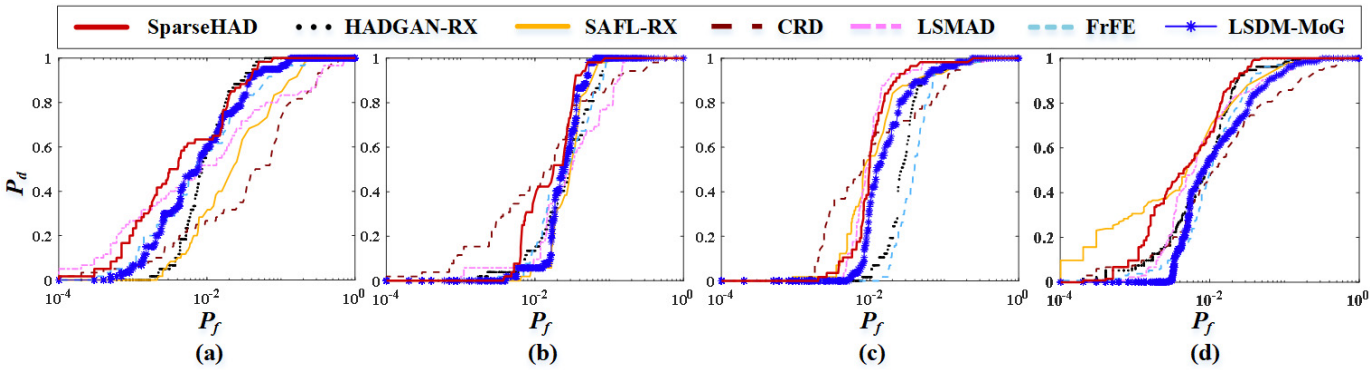


Fig. 6. ROC curves of (P_d, P_f) for the methods on (a) Gulfport, (b) Gainesville, (c) San Diego-1, and (d) San Diego-2.

TABLE V
EVALUATION AUC SCORES OF THE DIFFERENT NETWORK-BASED METHODS ON DIFFERENT DATASETS

Datasets	The AUC scores of $(P_d, P_f)/(P_f, \tau)$						
	sparseHAD	HADGAN_RX	SFAL_RX	CRD	LSMAD	FrFE	LSDM_MoG
G1	**0.9886**/0.0185	0.9870/0.0765	0.8983/0.0323	0.9239/0.0324	0.9762/0.0213	0.9828/0.1135	0.9521/**0.0037**
G2	**0.9792**/0.0052	0.9661/0.0419	0.9533/0.0216	0.9493/0.0254	0.9649/0.0356	0.9728/0.0272	0.9577/**0.0039**
S1	**0.9846**/0.0101	0.9644/0.0450	0.9655/0.1526	0.9816/0.0089	0.9534/0.0369	0.9750/0.0231	0.9789/**0.0026**
S2	**0.9914**/0.0140	0.9860/0.0339	0.9444/0.0742	0.9792/0.0124	0.9832/0.0461	0.9742/0.0177	0.9893/**0.0016**
Average	**0.9860**/0.0120	0.9759/0.0493	0.9403/0.0701	0.9585/0.0198	0.9694/0.0349	0.9762/0.0453	0.9695/**0.0029**

TABLE VI
AVERAGE COMPUTING TIME (IN SECONDS) OF THE COMPARATIVE METHODS

Datasets	sparseHAD	HADGAN_RX	SFAL_RX	CRD	LSMAD	FrFE	LSDM_MoG
G1	5.3900	**0.2646**	0.4689	98.185	27.875	20.738	43.123
G2	5.4570	**0.2454**	0.4446	45.794	19.394	17.436	45.794
S1	5.4405	**0.2486**	0.4320	73.576	17.567	22.017	24.145
S2	5.4763	**0.2499**	0.4889	99.411	25.642	17.976	37.912
Average	5.4409	**0.2521**	0.4586	79.241	22.619	19.541	37.743

advantage over other comparisons. Given that HAD focuses on whether anomalies can be detected, the AUC score of (P_f, τ) fits perfectly with HAD. In summary, the AUC score in Table V and the ROC curve in Fig. 6 are consistent with the visual detection maps in Fig. 5, so it can be concluded that the sparseHAD method is competitive with other advanced methods.

4) Computing Time: The average computing time of each comparison method on all the test images is recorded in seconds in Table VI. The deep learning-based methods, sparseHAD, HADGAN_RX, SFAL_RX, are far more efficient than the traditional methods, CRD, LSMAD, FrFE, and LSDM_MoG. This is because the test images are fed to a highly efficient RX detector directly after being trained on the network. Due to the influence of local window parameters, the running time of CRD method is much longer than other methods. Although the proposed method is more time-consuming than HADGAN_RX and SAEL_RX methods, it is still acceptable for HAD. Furthermore, time consumption will likely continue to decrease as more efficient computing resources are used in the future. The above analysis again confirms the powerful competitiveness of the sparseHAD.

VI. CONCLUSION

In this article, we study the HAD task from a novel weakly supervised perspective where a spectral mapping model is formed to reconstruct background samples with small errors. The proposed sparseHAD examines the role of three subnetworks, especially the sparseNet, and reformulates SC into the objective function of the other two GAN-based subnetworks. Furthermore, we dive deep into the mechanics of reconstruction error calculation in different spaces. The empirical findings in the assessment experiment provide an insight into the robustness of the proposed method. The ablation study further ensures its reliability and effectiveness. Experiments of performance comparison show the sparseHAD method outperforms state-of-the-art HAD methods. The superior performance of sparseHAD exhibits the value of WSL and SC in HSI analysis.

REFERENCES

[1] T. Schlegl, P. Seeböck, S. M. Waldstein, G. Langs, and U. Schmidt-Erfurth, "f-AnoGAN: Fast unsupervised anomaly detection with generative adversarial networks," *Med. Image Anal.*, vol. 54, pp. 30–44, May 2019.

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[2] S. Akcay, A. Atapour-Abarghouei, and T. P. Breckon, "Ganomaly: Semi-supervised anomaly detection via adversarial training," in *Proc. Asian Conf. Comput. Vis.*, 2018, pp. 622–637.

[3] T. Jiang, Y. Li, W. Xie, and Q. Du, "Discriminative reconstruction constrained generative adversarial network for hyperspectral anomaly detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 7, pp. 4666–4679, Jul. 2020.

[4] B. A. Olshausen and D. J. Field, "Emergence of simple-cell receptive field properties by learning a sparse code for natural images," *Nature*, vol. 381, no. 6583, pp. 607–609, 1996.

[5] B. A. Olshausen and D. J. Field, "Sparse coding with an overcomplete basis set: A strategy employed by v1?" *Vis. Res.*, vol. 37, no. 23, pp. 3311–3325, Dec. 1997.

[6] Y. Cong, J. Yuan, and J. Liu, "Sparse reconstruction cost for abnormal event detection," in *Proc. IEEE Conf. Comput. Vis. Pattern Recog.*, Jun. 2011, pp. 3449–3456.

[7] C. Lu, J. Shi, and J. Jia, "Abnormal event detection at 150 FPS in MATLAB," in *Proc. IEEE Int. Conf. Comput. Vis.*, Dec. 2013, pp. 2720–2727.

[8] S. Arora, R. Ge, T. Ma, and A. Moitra, "Simple, efficient, and neural algorithms for sparse coding," in *Proc. Conf. Learn. Theory*, Jun. 2015, pp. 113–149.

[9] F. Xiong, J. Zhou, and Y. Qian, "Material based object tracking in hyperspectral videos," *IEEE Trans. Image Process.*, vol. 29, pp. 3719–3733, 2020.

[10] A. F. H. Goetz, G. Vane, J. E. Solomon, and B. N. Rock, "Imaging spectrometry for earth remote sensing," *Science*, vol. 228, no. 4704, pp. 1147–1153, 1985.

[11] A. Plaza *et al.*, "Recent advances in techniques for hyperspectral image processing," *Remote Sens. Environ.*, vol. 113, pp. S110–S122, Sep. 2009.

[12] Q. Wang, J. Lin, and Y. Yuan, "Salient band selection for hyperspectral image classification via manifold ranking," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 27, no. 6, pp. 1279–1289, Jun. 2016.

[13] N. M. Nasrabadi, "Hyperspectral target detection: An overview of current and future challenges," *IEEE Signal Process. Mag.*, vol. 31, no. 1, pp. 34–44, Jan. 2014.

[14] L. Li, W. Li, Q. Du, and R. Tao, "Low-rank and sparse decomposition with mixture of Gaussian for hyperspectral anomaly detection," *IEEE Trans. Cybern.*, early access, Feb. 25, 2020, doi: 10.1109/TCYB.2020.2968750.

[15] P. Ghamisi *et al.*, "Advances in hyperspectral image and signal processing: A comprehensive overview of the state of the art," *IEEE Geosci. Remote Sens. Mag.*, vol. 5, no. 4, pp. 37–78, Dec. 2017.

[16] Z.-H. Zhou, "A brief introduction to weakly supervised learning," *Nat. Sci. Rev.*, vol. 5, no. 1, pp. 44–53, 2018.

[17] T. Ergen and S. S. Kozat, "Unsupervised anomaly detection with LSTM neural networks," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 31, no. 8, pp. 3127–3141, Aug. 2019.

[18] K. Zhou *et al.*, "Sparse-GAN: Sparsity-constrained generative adversarial network for anomaly detection in retinal OCT image," in *Proc. IEEE 17th Int. Symp. Biomed. Imag. (ISBI)*, Apr. 2020, pp. 1227–1231.

[19] W. Luo, W. Liu, and S. Gao, "A revisit of sparse coding based anomaly detection in stacked RNN framework," in *Proc. IEEE Int. Conf. Comput. Vis. (ICCV)*, Oct. 2017, pp. 341–349.

[20] I. S. Reed and X. Yu, "Adaptive multiple-band CFAR detection of an optical pattern with unknown spectral distribution," *IEEE Trans. Acoust., Speech Signal Process.*, vol. 38, no. 10, pp. 1760–1770, Oct. 1990.

[21] J. M. Molero, E. M. Garzón, I. García, and A. Plaza, "Analysis and optimizations of global and local versions of the RX algorithm for anomaly detection in hyperspectral data," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 6, no. 2, pp. 801–814, Apr. 2013.

[22] H. Kwon and N. M. Nasrabadi, "Kernel RX-algorithm: A nonlinear anomaly detector for hyperspectral imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 43, no. 2, pp. 388–397, Feb. 2005.

[23] J. Li, H. Zhang, L. Zhang, and L. Ma, "Hyperspectral anomaly detection by the use of background joint sparse representation," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 8, no. 6, pp. 2523–2533, Jun. 2015.

[24] L. Wei and D. Qian, "Collaborative representation for hyperspectral anomaly detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 53, no. 3, pp. 1463–1474, Mar. 2015.

[25] Y. Zhang, B. Du, L. Zhang, and S. Wang, "A low-rank and sparse matrix decomposition-based Mahalanobis distance method for hyperspectral anomaly detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 54, no. 3, pp. 1376–1389, Mar. 2016.

[26] Y. Qu *et al.*, "Anomaly detection in hyperspectral images through spectral unmixing and low rank decomposition," in *Proc. IEEE Int. Geosci. Remote Sens. Symp. (IGARSS)*, Jul. 2016, pp. 1855–1858.

[27] Y. Qu *et al.*, "Hyperspectral anomaly detection through spectral unmixing and dictionary-based low-rank decomposition," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 8, pp. 4391–4405, Aug. 2018.

[28] T. Cheng and B. Wang, "Graph and total variation regularized low-rank representation for hyperspectral anomaly detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 1, pp. 391–406, Jan. 2020.

[29] S. Chang, B. Du, and L. Zhang, "A subspace selection-based discriminative forest method for hyperspectral anomaly detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 6, pp. 4033–4046, Jun. 2020.

[30] X. Zhang, X. Ma, N. Huyan, J. Gu, X. Tang, and L. Jiao, "Spectral-difference low-rank representation learning for hyperspectral anomaly detection," *IEEE Trans. Geosci. Remote Sens.*, early access, Jan. 13, 2021, doi: 10.1109/TGRS.2020.3046727.

[31] L. Li, W. Li, Y. Qu, C. Zhao, R. Tao, and Q. Du, "Prior-based tensor approximation for anomaly detection in hyperspectral imagery," *IEEE Trans. Neural Netw. Learn. Syst.*, early access, Dec. 9, 2021, doi: 10.1109/TNNLS.2020.3038659.

[32] X. Kang, X. Zhang, S. Li, K. Li, J. Li, and J. A. Benediktsson, "Hyperspectral anomaly detection with attribute and edge-preserving filters," *IEEE Trans. Geosci. Remote Sens.*, vol. 55, no. 10, pp. 5600–5611, Oct. 2017.

[33] W. Xie, T. Jiang, Y. Li, X. Jia, and J. Lei, "Structure tensor and guided filtering-based algorithm for hyperspectral anomaly detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 7, pp. 4218–4230, Jul. 2019.

[34] R. Tao, X. Zhao, W. Li, H. Li, and Q. Du, "Hyperspectral anomaly detection by fractional Fourier entropy," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 12, no. 12, pp. 4920–4929, Dec. 2019.

[35] Z. Huang, L. Fang, and S. Li, "Subpixel-pixel-superpixel guided fusion for hyperspectral anomaly detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 9, pp. 5998–6007, Sep. 2020.

[36] V. L. Cao, M. Nicolau, and J. McDermott, "Learning neural representations for network anomaly detection," *IEEE Trans. Cybern.*, vol. 49, no. 8, pp. 3074–3087, Aug. 2019.

[37] K. Jiang, W. Xie, Y. Li, J. Lei, G. He, and Q. Du, "Semisupervised spectral learning with generative adversarial network for hyperspectral anomaly detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 7, pp. 5224–5236, Jul. 2020.

[38] W. Xie, B. Liu, Y. Li, J. Lei, and Q. Du, "Autoencoder and adversarial-learning-based semisupervised background estimation for hyperspectral anomaly detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 8, pp. 5416–5427, Aug. 2020.

[39] G. E. Hinton and R. S. Zemel, "Autoencoders, minimum description length and Helmholtz free energy," in *Proc. Adv. Neural Inf. Process. Syst.*, 1994, pp. 3–10.

[40] C. Zhao and L. Zhang, "Spectral-spatial stacked autoencoders based on low-rank and sparse matrix decomposition for hyperspectral anomaly detection," *Infr. Phys. Technol.*, vol. 92, pp. 166–176, Aug. 2018.

[41] W. Xie, J. Lei, B. Liu, Y. Li, and X. Jia, "Spectral constraint adversarial autoencoders approach to feature representation in hyperspectral anomaly detection," *Neural Netw.*, vol. 119, pp. 222–234, Nov. 2019.

[42] B. Hosseiny and R. Shah-Hosseini, "A hyperspectral anomaly detection framework based on segmentation and convolutional neural network algorithms," *Int. J. Remote Sens.*, vol. 41, no. 18, pp. 6946–6975, Sep. 2020.

[43] X. Lu, W. Zhang, and J. Huang, "Exploiting embedding manifold of autoencoders for hyperspectral anomaly detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 3, pp. 1527–1537, Mar. 2020.

[44] I. Goodfellow *et al.*, "Generative adversarial nets," in *Proc. Adv. Neural Inf. Process. Syst.*, 2014, pp. 2672–2680.

[45] W. Li, G. Wu, and Q. Du, "Transferred deep learning for anomaly detection in hyperspectral imagery," *IEEE Geosci. Remote Sens. Lett.*, vol. 14, no. 5, pp. 597–601, May 2017.

[46] S. Song, H. Zhou, Y. Yang, and J. Song, "Hyperspectral anomaly detection via convolutional neural network and low rank with density-based clustering," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 12, no. 9, pp. 3637–3649, Sep. 2019.

[47] X. Fu, S. Jia, L. Zhuang, M. Xu, J. Zhou, and Q. Li, "Hyperspectral anomaly detection via deep plug-and-play denoising CNN regularization," *IEEE Trans. Geosci. Remote Sens.*, early access, Jan. 20, 2021, doi: 10.1109/TGRS.2021.3049224.

[48] K. Jiang, W. Xie, J. Lei, T. Jiang, and Y. Li, "LREN: Low-rank embedded network for sample-free hyperspectral anomaly detection," in *Proc. AAAI Conf. Artif. Intell.*, vol. 35, 2020, pp. 4139–4146.

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[49] X. Peng, C. Lu, Z. Yi, and H. Tang, "Connections between nuclear-norm and Frobenius-norm-based representations," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 29, no. 1, pp. 218–224, Jan. 2018.
[50] S. Gu, W. Zuo, Q. Xie, D. Meng, X. Feng, and L. Zhang, "Convolutional sparse coding for image super-resolution," in *Proc. IEEE Int. Conf. Comput. Vis.*, 2015, pp. 1823–1831.
[51] W. Luo *et al.*, "Video anomaly detection with sparse coding inspired deep neural networks," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 43, no. 3, pp. 1070–1084, Mar. 2021.
[52] M. Ester, H. Kriegel, J. Sander, and X. Xu, "A density-based algorithm for discovering clusters in large spatial databases with noise," in *Proc. 2nd Int. Conf. Knowl. Discov. Data Mining*, 1996, pp. 226–231.
[53] P. Perera, R. Nallapati, and B. Xiang, "OCGAN: One-class novelty detection using GANs with constrained latent representations," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2019, pp. 2898–2906.
[54] S. Pidhorskyi, R. Almohsen, and G. Doretto, "Generative probabilistic novelty detection with adversarial autoencoders," in *Proc. Adv. Neural Inf. Process. Syst.*, 2018, pp. 6823–6834.
[55] J. T. Zhou *et al.*, "SC2Net: Sparse lstms for sparse coding," in *Proc. AAAI Conf. Artif. Intel.*, 2018, pp. 4588–4595.
[56] M. S. Lewicki and T. J. Sejnowski, "Learning overcomplete representations," *Neural Comput.*, vol. 12, no. 2, pp. 337–365, 2000.
[57] Y. Xu, Z. Wu, J. Li, A. Plaza, and Z. Wei, "Anomaly detection in hyperspectral images based on low-rank and sparse representation," *IEEE Trans. Geosci. Remote Sens.*, vol. 54, no. 4, pp. 1990–2000, Apr. 2016.
[58] W. Xie, B. Liu, Y. Li, J. Lei, C. Chang, and G. He, "Spectral adversarial feature learning for anomaly detection in hyperspectral imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 4, pp. 2352–2365, Apr. 2019.
[59] A. P. Bradley, "The use of the area under the ROC curve in the evaluation of machine learning algorithms," *Pattern Recognit.*, vol. 30, no. 7, pp. 1145–1159, 1997.
[60] D. P. Kingma and J. Ba, "Adam: A method for stochastic optimization," 2014, *arXiv:1412.6980*. [Online]. Available: http://arxiv.org/abs/1412.6980



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